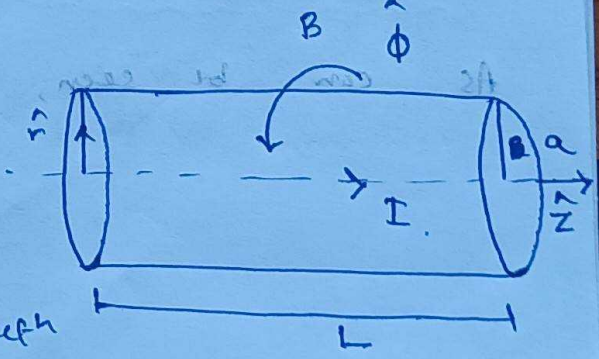
Assignment - 6

① A straight cylindrical wire of radius a, length L & resistance R carries a steady current I.

(a) Calculate the electric field $\bar{E}$ & magnetic field $\bar{B}$ @ the surface of the wire.

(b) Calculate the Poynting vector $\bar{S}$ @ the surface of the wire.

(c) Integrate the Poynting vector over the surface of the wire to find the total power flowing into the conductor. Further, verify that it matches the expected Joule heating formula I^2R.

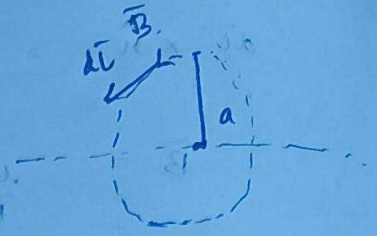


Ans: (a) $V = IR$ (across wire)

$$E = \frac{V}{L} = \frac{IR}{L}.$$

$$\boxed{\bar{E} = \frac{IR}{L}\hat{z}}$$ (in the same direction as ↓ V. as I)

From Ampere's law:



Amperian Loop.

$$\oint \bar{B} \cdot d\bar{l} = \mu_0 I$$

$$\Rightarrow \quad B \oint dl = \mu_0 I$$

$$\Rightarrow \quad B(2\pi a) = \mu_0 I$$

$$\Rightarrow \quad B = \frac{\mu_0 I}{2\pi a}$$

$$\boxed{\bar{B} = \frac{\mu_0 I}{2\pi a} \hat{\phi}}$$

(b) $$\bar{S} = \frac{1}{\mu_0} (\bar{E} \times \bar{B})$$

$$= \frac{1}{\mu_0} \left[\frac{IR}{L} \hat{z} \times \frac{\mu_0 I}{2\pi a} \hat{\phi} \right]$$

$$\boxed{\bar{S} = \frac{I^2 R}{2\pi a L} (-\hat{r})} \qquad [\hat{z} \times \hat{\phi} = -\hat{r}]$$

(c) The -ve sign indicates that ~~power~~ energy flows into the wire from the surroundings, where it is dissipated as heat.

(d) $$P = \oiint \bar{S} \cdot d\bar{A}$$

$$d\bar{A} = (a\, d\phi)(dz)\, \hat{r}$$

$$\bar{S} \cdot d\bar{A} = -\frac{I^2 R}{2\pi a L} \hat{r} \cdot a\, d\phi\, dz\, \hat{r} = -\frac{I^2 R}{2\pi L} d\phi\, dz$$

$$P = \int \bar{S} \cdot d\bar{A} = \frac{-I^2 R}{2\pi L} \int_0^{2\pi} d\phi \int_0^L dz = \boxed{-I^2 R = P}$$

The -ve sign indicates that power flows into the wire from the surroundings.

As can be seen, it matches the result for Joule's heating.

(2) A long straight ~~wire~~ co-axial cable consists of an inner cylindrical conductor of radius a & a thin outer cylindrical conducting shell of radius b. A DC battery maintains a steady potential difference V b/w the conductors & a steady ~~potential difference~~ current I flows down the inner conductor and returns via the outer conductor. Assume that the space b/w the conductors is vacuum.

~~Ans~~ (a) Determine the electric field $\bar{E}$ & magnetic field $\bar{B}$ in the region $a < r < b$, where r is the radial ~~direction~~ distance from the central axis.

(b) Calculate the Poynting vector $\bar{S}$ in the given region.

(c) Integrate the poynting vector over the cross-sectional area b/w the conductors to find the total power P transmitted down the cable & verify that it matches the standard expression $P = VI$

Ans: (a) ~~For determin~~ $E = -\nabla V$

For finding out V, we use the Laplace's eqn in the volume b/w the cylinders.

$$\nabla^2 V = 0 \quad [\text{vacuum b/w cylinders}; \rho = 0]$$

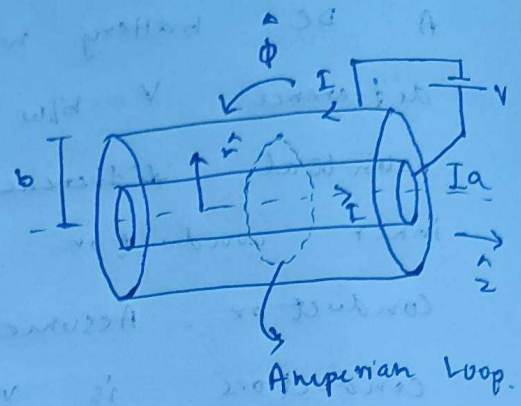
~~In cylindrical coordinates~~, Since there is cylindrical symmetry, so no ϕ & z dependence

$$\nabla^2 V \Rightarrow \frac{1}{r}\frac{d}{dr}\left(r\frac{dV}{dr}\right) = 0$$

$$\Rightarrow \frac{d}{dr}\left(r\frac{dV}{dr}\right) = 0$$

$\Rightarrow r\frac{dV}{dr} = c_1$

$\Rightarrow \frac{dV}{dr} = \frac{c_1}{r} \Rightarrow E(r) = -\frac{dV}{dr} = -\frac{c_1}{r}$



Applying BCs :

$V(b) = 0 \; ; \; V(a) = V$

$\therefore \frac{dV}{dr} = \frac{c_1}{r}$

$\Rightarrow V = c_1 \int \frac{dr}{r}$

$\Rightarrow \boxed{V(r) = c_1 \ln r + c_2}$

Using BCs : $V(a) = c_1 \ln a + c_2 = V$

$V(b) \Rightarrow c_1 \ln b + c_2 = 0$

$\Rightarrow \boxed{c_2 = -c_1 \ln b}$

$\therefore c_1 \ln a - c_1 \ln b = V$

$\Rightarrow c_1 = \frac{V}{\cancel{\ln(b/a)}} \; \frac{V}{\ln(a/b)}$

$\therefore V(r) = \frac{V \ln r}{\ln(\frac{a}{b})} + - \frac{V}{\ln(\frac{a}{b})} \ln b$

$\boxed{V(r) = \frac{V}{\ln(\frac{a}{b})} \ln\left(\frac{r}{b}\right)}$

$\therefore E(r) = -\frac{dV}{dr} = -\frac{c_1}{r} = -\frac{V}{r \ln(\frac{a}{b})} = \frac{V}{r \ln(\frac{b}{a})}$

$\therefore \boxed{\bar{E}(r) = \frac{V}{r \ln(\frac{b}{a})} \hat{r}}$ (Ans) (direction is same as direct^n of ~~I = radial~~ $\downarrow$ V).

From Ampere's Law :

$\oint \bar{B} \cdot d\bar{l} = \mu_0 I$ (for the given Amperian loop)

$\Rightarrow B \oint dl = \mu_0 I$

$\Rightarrow B = \frac{\mu_0 I}{2\pi r} \Rightarrow \boxed{\bar{B}(r) = \frac{\mu_0 I}{2\pi r} \hat{\phi}}$ (Ans)

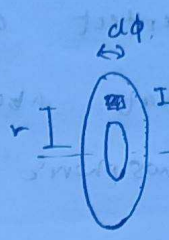
(b) $\bar{S} = \frac{1}{\mu_0}(\bar{E} \times \bar{B})$;

$$= \frac{1}{\cancel{\mu_0}} \frac{V}{r \ln\left(\frac{b}{a}\right)} \cdot \frac{\cancel{\mu_0} I}{2\pi r} \cdot (\hat{r} \times \hat{\phi})$$

$$\boxed{\bar{S} = \frac{VI}{2\pi r^2} \cdot \frac{1}{\ln\left(\frac{b}{a}\right)} \hat{z}} \quad \text{(Ans)}$$

$\bar{S}$ flows b/w the conductors, not inside. It transfers energy from one end to the other.

(c) $P = \int \bar{S} \cdot d\bar{A}$



$dA = r\, dr\, d\phi$.

$d\bar{A} = r\, dr\, d\phi\, \hat{z}$.

$$\int \bar{S} \cdot d\bar{A} = \frac{VI}{2\pi r} \cdot \frac{1}{\ln(b/a)} \int_a^b \frac{dr}{r} \int_0^{2\pi} d\phi$$

$$= \frac{VI}{2\pi \ln(b/a)} \ln r \Big|_a^b \cdot 2\pi$$

$$= \frac{VI}{\ln(b/a)} \cdot \ln(b/a) = \boxed{VI = P} \quad \text{(Approved)}$$

It matches the familiar expression of $P = VI$.

③ Show that in vacuum, magnetic field satisfies the eq^n $\mu_0 \epsilon_0 \frac{\partial^2 \bar{B}}{\partial t^2} = \nabla^2 \bar{B}$

Ans: Maxwell's eq^n in vacuum:

(i) $\bar{\nabla} \cdot \bar{E} = 0$ (ii) $\bar{\nabla} \times \bar{E} = -\frac{\partial \bar{B}}{\partial t}$

(iii) $\bar{\nabla} \cdot \bar{B} = 0$ (iv) $\bar{\nabla} \times \bar{B} = \mu_0 \epsilon_0 \frac{\partial \bar{E}}{\partial t}$

Taking curl of eq^n (iv):

$$\bar{\nabla} \times (\bar{\nabla} \times \bar{B}) = \bar{\nabla} \times \left(\mu_0 \epsilon_0 \frac{\partial \bar{E}}{\partial t}\right)$$

$$\Rightarrow (\bar{\nabla} \cdot \bar{B})\bar{\nabla} - (\bar{\nabla} \cdot \bar{\nabla})\bar{B} = \mu_0 \epsilon_0 \frac{\partial}{\partial t}(\bar{\nabla} \times \bar{E})$$

[from vector identity]

$$\Rightarrow -\nabla^2 \bar{B} = \mu_0 \epsilon_0 \frac{\partial}{\partial t}\left(-\frac{\partial \bar{B}}{\partial t}\right)$$ [from (ii) & (iii)]

$$\Rightarrow \boxed{\mu_0 \epsilon_0 \frac{\partial^2 \bar{B}}{\partial t^2} = \nabla^2 \bar{B}}$$ (Proved)